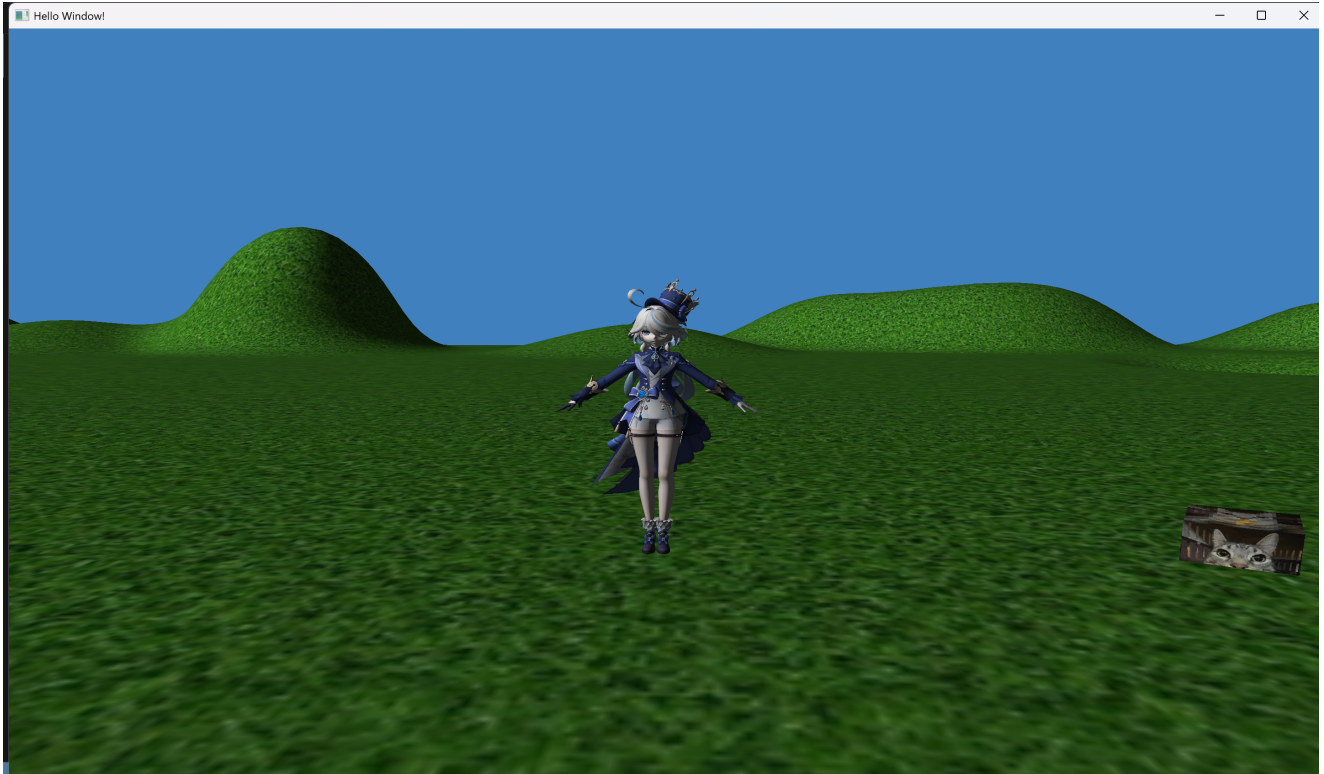


OpenGL Assignment

This project is a 3D OpenGL application built with CMake, GLFW, GLAD, and GLM. It uses a simple component-system approach to manage objects, camera, physics, and rendering.

Preview



Features

- 3D rendering with the OpenGL core profile
- FPS-style camera with mouse look
- Camera movement with **W, A, S, D**
- Jump with **Space**
- Exit the application with **Esc**
- Render multiple objects and models from the **models/** folder

Project Structure

- **src/** - main source code
- **src/components/** - entity components
- **src/systems/** - systems for motion, collision, camera, and rendering
- **src/factories/** - entity and mesh creation
- **src/view/** - shader and rendering utilities
- **src/shaders/** - vertex and fragment shader files
- **dependencies/** - third-party library headers
- **models/** - model and texture assets

Requirements

- CMake 3.12 or newer
- A C++20 compiler
- OpenGL
- GLFW

GLAD and GLM are already included in this repository.

Build Instructions

Run the following from the project root:

```
cmake -S . -B build
cmake --build build
```

If your generator or toolchain is different, adjust the CMake commands to match your environment.

Run Instructions

After the build succeeds, run the generated executable named `hello_window` from the build output folder.

Example:

```
./build/hello_window
```

On Windows, the executable may be named `hello_window.exe`.

Controls

- `W` - move forward
- `A` - move left
- `S` - move backward
- `D` - move right
- `Mouse` - change the camera direction
- `Space` - jump
- `Esc` - quit

Notes